SOCIOL 723: Social Statistics II

Week 4, Maximum Likelihood: Applications

Wenhao Jiang
Department of Sociology, Fall 2026

Duke UNIVERSITY

Today

The Generalized Linear Model

Binary Outcomes

Interpretation

Multinomial, Ordered, and Count Outcomes

Event History (Survival) Analysis

Diagnostics and Practice

★ = essential, you must understand this ★ = for understanding, not examined

The GLM

Putting Covariates Inside the Likelihood ★

Last week the parameter θ was a single number. Now we let it depend on covariates. Three ingredients:

The GLM recipe

1. **Random component**: a distribution for Y_i with mean $\mu_i = E[Y_i \mid X_i]$, normal, Bernoulli, Poisson, ...
2. **Systematic component**: a *linear predictor* $\eta_i = X_i'\boldsymbol{\beta}$.
3. **Link function**: $g(\mu_i) = \eta_i$, connecting the two.

- The linear predictor is always linear. All the nonlinearity lives in the link, whose job is to map μ_i from its natural range (a probability in $[0, 1]$, a rate in $[0, \infty)$) onto the whole real line.
- This is why we can keep using everything from Weeks 1–2 about $X'\beta$.

The Standard Family

Outcome	Distribution	Link $g(\mu)$	R
Continuous	Normal	identity: μ	`glm(..., gaussian)` = `lm`
Binary	Bernoulli	logit: $\log \frac{\mu}{1-\mu}$	`binomial(link="logit")`
Binary	Bernoulli	probit: $\Phi^{-1}(\mu)$	`binomial(link="probit")`
Count	Poisson	log: $\log \mu$	`poisson(link="log")`
Count, overdispersed	Neg. binomial	log	`MASS::glm.nb`
Nominal, J categories	Multinomial	multi-logit	`nnet::multinom`
Ordered, J categories	latent logistic	cumulative logit	`MASS::polr`

The *canonical* link for each distribution (logit for the binomial, log for the Poisson) makes the algebra especially clean, but you are free to choose another, probit is not canonical and nobody minds.

How `glm()` Actually Fits ★

Every GLM log-likelihood has score equations of the form

$$\sum_{i=1}^{n} \frac{\partial \mu_i}{\partial \eta_i} \frac{Y_i - \mu_i}{V(\mu_i)} X_i = \mathbf{0},$$

where $V(\mu)$ is the variance function implied by the distribution.

- Compare with OLS's normal equations $\sum_i X_i(Y_i - X_i'\boldsymbol{\beta}) = \mathbf{0}$: the difference is the weight $\frac{\partial\mu/\partial\eta}{V(\mu)}$.
- So GLMs are fitted by iteratively reweighted least squares: run a weighted regression, update the weights from the new fitted values, repeat. This is Fisher scoring in disguise.
- Practical consequence: `glm()` usually converges in 4–6 iterations, and failure to converge is a real signal (often separation).

Binary

Start With the Linear Probability Model ★

Just run OLS with a 0/1 outcome: $\Pr(Y_i = 1 \mid X_i) = X_i'\beta$.

In its favour

- Coefficients are marginal effects, directly on the probability scale.
- Everything from Weeks 1–2 applies: FWL, fixed effects, IV, clustering.
- Robust to functional form in the sense of being the best linear approximation to the CEF.

Against

- Fitted probabilities can fall outside $[0, 1]$.
- Errors are *necessarily* heteroskedastic: $\mathrm{Var}(Y_i \mid X_i) = p_i(1 - p_i)$. (Use robust SEs, Week 2.)
- Constant marginal effects are implausible near 0 and 1.

A reasonable position

For a causal contrast with a binary treatment, the LPM is often fine and easier to interpret. For prediction, extrapolation, or when the probabilities are near the boundary, you want logit or probit.

A Latent-Variable Motivation ★

Imagine an unobserved continuous propensity Y_i^*, a taste for voting, a level of labour-market attachment:

$$Y_i^* = X_i'\boldsymbol{\beta} + \epsilon_i, \qquad Y_i = \begin{cases} 1 & \text{if } Y_i^* > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$\Pr(Y_i = 1 \mid X_i) = \Pr(\epsilon_i > -X_i'\boldsymbol{\beta}) = F(X_i'\boldsymbol{\beta})$$

for F the CDF of $-\epsilon_i$.

- $\epsilon_i \sim$ **logistic** $\Rightarrow F(z) = \frac{e^z}{1+e^z} = \Lambda(z)$: the logit model.
- $\epsilon_i \sim \mathcal{N}(0,1) \Rightarrow F(z) = \Phi(z)$: the probit model.

The choice between them is a choice about the tails of an unobservable. It almost never matters empirically.

The Logit Likelihood ★

With $p_i \equiv \Lambda(X_i'\boldsymbol{\beta}) = \dfrac{\exp(X_i'\boldsymbol{\beta})}{1 + \exp(X_i'\boldsymbol{\beta})}$, the Bernoulli log-likelihood from last week becomes

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big[Y_i \log p_i + (1 - Y_i) \log(1 - p_i) \Big].$$

The Logit Likelihood ★

With $p_i \equiv \Lambda(X_i'\boldsymbol{\beta}) = \dfrac{\exp(X_i'\boldsymbol{\beta})}{1+\exp(X_i'\boldsymbol{\beta})}$, the Bernoulli log-likelihood from last week becomes

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big[Y_i \log p_i + (1-Y_i) \log(1-p_i) \Big].$$

Substituting and simplifying (using $\log \frac{p_i}{1-p_i} = X_i'\boldsymbol{\beta}$ and $\log(1-p_i) = -\log(1+e^{X_i'\boldsymbol{\beta}})$):

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big[Y_i\, X_i'\boldsymbol{\beta} - \log\left(1 + e^{X_i'\boldsymbol{\beta}}\right) \Big].$$

This is globally concave in $\boldsymbol{\beta}$, and *strictly* concave once $\boldsymbol{X}$ has full column rank, which is what makes the maximizer unique when it exists. Probit shares the property.

"When it exists" is doing real work. Concavity alone does not deliver a finite maximizer: under perfect separation the likelihood keeps climbing and $\hat{\boldsymbol{\beta}}$ diverges. We return to this later.

The Logit Score

Differentiate. The key fact is $\dfrac{\partial \Lambda(z)}{\partial z} = \Lambda(z)(1-\Lambda(z))$:

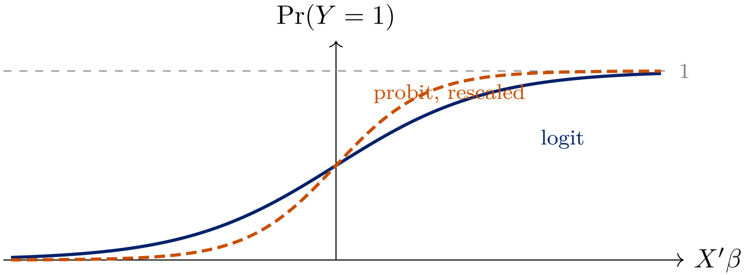
$$\frac{\partial \ell}{\partial \boldsymbol{\beta}} = \sum_{i=1}^{n} \left[Y_i X_i - \frac{e^{X_i'\boldsymbol{\beta}}}{1+e^{X_i'\boldsymbol{\beta}}} X_i \right] = \sum_{i=1}^{n} (Y_i - p_i) X_i \stackrel{!}{=} \mathbf{0}.$$

Look at what this says

$\sum_i (Y_i - \hat{p}_i) X_i = \mathbf{0}$: the residuals $Y_i - \hat{p}_i$ are orthogonal to the regressors. Exactly the OLS normal equations, with the fitted probability in place of the fitted value.

There is no closed-form solution because p_i depends on $\boldsymbol{\beta}$ nonlinearly, hence the numerical optimization from last week. The information matrix ★ is $\mathcal{I}(\boldsymbol{\beta}) = \sum_i p_i(1-p_i) X_i X_i'$.

Logit Versus Probit



(The sketched "probit" curve is the standard logistic approximation $\Lambda(1.702x)$ to $\Phi(x)$; the two agree to within 0.01 everywhere, which is itself the point of the slide.)

- The two curves are nearly indistinguishable after rescaling: logit coefficients are roughly 1.6–1.8 times probit coefficients.

Logit Versus Probit (cont.)

- Choose logit if you want odds ratios; probit if you want a normal latent variable (useful for bivariate probit and selection models).
- Marginal effects and predicted probabilities from the two are essentially identical. Do not agonize over this choice.

The Scale Is Not Identified, and Why It Matters ★

In the latent-variable setup, only β/σ_ϵ is identified: we normalize σ_ϵ (to $\pi/\sqrt{3}$ for logit, to 1 for probit) because the data cannot tell us anything about it.

The consequence (Allison 1999; Mood 2010)

Adding a covariate changes the residual variance and therefore *rescales all the coefficients*, even ones whose true effect is unchanged:

- You cannot compare logit coefficients across nested models the way you compare OLS coefficients.
- You cannot compare logit coefficients across groups or samples with different residual variance.
- "The coefficient shrank when I added controls" may be pure rescaling rather than confounding.

The fix: compare *marginal effects* or *predicted probabilities*, which are on the probability scale and do not suffer from this.

Interpretation

Three Scales, Three Meanings ★

Scale	Quantity	Interpretation of β_j
Log-odds	$\log \frac{p}{1-p} = X'\boldsymbol{\beta}$	additive: linear, but nobody thinks in log-odds
Odds	$\frac{p}{1-p} = e^{X'\boldsymbol{\beta}}$	multiplicative: e^{β_j} is the *odds ratio*
Probability	$p = \Lambda(X'\boldsymbol{\beta})$	what we care about, but the effect is not constant

- The model is linear on the first scale and nonlinear on the third. That tension is the whole interpretation problem.
- An odds ratio is not a risk ratio, and it is not a probability difference. An odds ratio of 2 can correspond to a probability change from 0.01 to 0.02 or from 0.33 to 0.50.
- Report probabilities or marginal effects. Report odds ratios only if your audience genuinely works in odds.

Marginal Effects ★

For a *continuous* X_j, the marginal effect on the probability is

$$\frac{\partial \Pr(Y_i = 1 \mid X_i)}{\partial X_{ij}} = \underbrace{\lambda(X_i'\beta)}_{\text{density at } X_i} \; \beta_j, \qquad \lambda(z) = \Lambda(z)(1 - \Lambda(z)).$$

For a *binary or categorical* X_j a derivative is meaningless; report the average *discrete change* $\frac{1}{n}\sum_i [\hat{p}_i(X_j{=}1) - \hat{p}_i(X_j{=}0)]$ instead. This is what `avg_comparisons()` does automatically. It depends on X_i, so there is no single "the" marginal effect. Two standard summaries:

- **Marginal effect at the mean (MEM)**: $\lambda(\bar{X}'\hat{\beta})\,\hat{\beta}_j$. Fast, but evaluated at a hypothetical person who may not exist (mean of a dummy = 0.53 female).
- **Average marginal effect (AME)**: $\frac{1}{n}\sum_i \lambda(X_i'\hat{\beta})\,\hat{\beta}_j$. Computed for every real respondent, then averaged. This is the default you should use.

Hanmer and Kalkan (2013) make the case for the AME ("observed-value") approach over the MEM ("average-case") approach.

Predicted Probabilities Done Right ★

Most readers want: *what is the predicted probability at $X_j = a$ versus $X_j = b$, holding everything else at realistic values?*

The observed-value approach (Hanmer and Kalkan 2013)

1. Take the whole sample as it is.
2. Set $X_j = a$ for *everyone*, leaving all other covariates at their actual values. Compute $\hat{p}_i$ for each i and average.
3. Repeat with $X_j = b$.
4. Report both, and their difference.

- This targets a population-level quantity, and is exactly the *g-computation* estimator we meet again in Week 6.
- Contrast with fixing every other covariate at its mean, which describes a person who may not exist.
- R: `marginaleffects::avg_predictions()`, `avg_slopes()`, `avg_comparisons()`.

Uncertainty for These Quantities

Predicted probabilities and marginal effects are *nonlinear* functions of $\hat{\beta}$, so you cannot read their standard errors off the regression table. Three options:

1. **Delta method** (Week 2): linearize and use $\nabla g(\hat{\beta})' \hat{V}(\hat{\beta}) \nabla g(\hat{\beta})$. Fast, analytic, and the default in `marginaleffects`.
2. **Simulation** (King, Tomz, and Wittenberg 2000): draw $\tilde{\beta}^{(s)} \sim \mathcal{N}(\hat{\beta}, \hat{V}(\hat{\beta}))$ for $s = 1, \ldots, S$, compute the quantity of interest at each draw, and take percentiles. Trivial to implement, handles any nonlinearity.
3. **Bootstrap**: resample the data, refit, recompute. Slowest, makes the fewest assumptions about the shape of the sampling distribution.

Do not compute a confidence interval on the log-odds scale and then transform its endpoints if what you want is an interval for a *difference* in probabilities.

Interactions in Nonlinear Models

In a logit, the effect of X_1 on the probability depends on X_2 *even with no product term in the model*: the curve is steepest in the middle and flat at the ends.

- So a significant $\hat{\beta}_{12}$ on the log-odds scale is neither necessary nor sufficient for interaction on the probability scale (Ai and Norton 2003).
- Conversely, Berry, DeMeritt, and Esarey (2010) show that omitting the product term imposes a very specific, and usually unmotivated, form of interaction.
- **Practical advice**: include the product term if your theory says the latent-scale effects interact, and then report the interaction you care about as a difference in average marginal effects, with an interval.

`marginaleffects::avg_comparisons(m, variables = "x1", by = "x2")`.

More Outcomes

Multinomial Logit

For an unordered outcome with J categories, pick a base category J and model

$$\Pr(Y_i = j \mid X_i) = \frac{\exp(X_i'\boldsymbol{\beta}_j)}{1 + \sum_{m=1}^{J-1} \exp(X_i'\boldsymbol{\beta}_m)}, \qquad \boldsymbol{\beta}_J \equiv \mathbf{0}.$$

- You estimate $J - 1$ coefficient vectors, interpretation burden grows fast. Report predicted probabilities by category, not coefficient tables.
- Every coefficient is relative to the omitted base category. Changing the base changes every number without changing the model.
- **IIA**: the ratio $\Pr(Y = j)/\Pr(Y = m)$ does not depend on what other options exist, the "red bus / blue bus" problem. Testable in principle (Hausman–McFadden), but the tests behave poorly; think substantively instead.
- Alternatives when IIA is untenable: nested logit, multinomial probit, mixed logit.

Ordered Logit ★

For an ordered outcome, one latent variable and $J-1$ cutpoints:

$$Y_i^* = X_i'\boldsymbol{\beta} + \epsilon_i, \qquad Y_i = j \;\text{ if }\; \tau_{j-1} < Y_i^* \le \tau_j.$$

This gives the cumulative-logit form $\Pr(Y_i \le j \mid X_i) = \Lambda(\tau_j - X_i'\boldsymbol{\beta})$.

- Only *one* $\boldsymbol{\beta}$ regardless of J: far more parsimonious than multinomial logit. That parsimony is the proportional odds assumption, the effect of X is the same at every cutpoint.
- Test it with a Brant test, or by comparing to a generalized ordered logit that relaxes it (`VGAM`, `ordinal::clm`).
- If proportional odds fails badly and the categories are genuinely ordered, a partial proportional odds model is usually the right compromise.

Poisson Regression for Counts ★

$$Y_i \mid X_i \sim \text{Poisson}(\mu_i), \qquad \log \mu_i = X_i' \boldsymbol{\beta}.$$

- Log link keeps $\mu_i > 0$; coefficients are *semi-elasticities*: e^{β_j} is the multiplicative change in the expected count, so $\beta_j = 0.10$ means about 10% more events.
- **Exposure**: if units are observed for different lengths of time or have different populations at risk, model the *rate* by adding an `offset(log(exposure))`, a coefficient constrained to 1.
- The Poisson imposes $\text{Var}(Y_i \mid X_i) = E[Y_i \mid X_i]$. That is a strong restriction and it is usually violated.

Useful fact: with a log link, Poisson regression is *consistent for the conditional mean even when the Poisson distribution is wrong*, provided the mean is correctly specified. Pair it with robust standard errors and it is a workhorse for non-negative outcomes.

Overdispersion ★

In practice $\mathrm{Var}(Y_i \mid X_i) > E[Y_i \mid X_i]$ almost always, unobserved heterogeneity makes some units systematically more event-prone.

- Consequence: coefficients stay roughly right, but standard errors are far too small, sometimes by a factor of two or three.
- **Quasi-Poisson**: keep the Poisson mean, multiply the variance by a dispersion parameter ϕ estimated from the data. Simple; inflates the standard errors by $\sqrt{\hat{\phi}}$.
- **Negative binomial**: model the heterogeneity explicitly as a gamma mixture, giving $\mathrm{Var}(Y) = \mu + \mu^2/\theta$. A genuine likelihood, so AIC and LR tests apply. `MASS::glm.nb`.
- **Robust standard errors** on a Poisson fit are the pragmatic third option, and are consistent under any variance function.

Test with a score test for overdispersion (`AER::dispersiontest`), a nice example of a test that needs only the restricted model.

Excess Zeros

Many sociological counts have far more zeros than any Poisson or negative binomial allows: number of arrests, number of union grievances, hours of volunteering.

- **Zero-inflated**: a mixture of a "structural zero" class (people who could never have a positive count) and a count process. Two equations, two sets of coefficients. `pscl::zeroinfl`.
- **Hurdle**: a binary model for zero versus positive, then a truncated count model for the positives. `pscl::hurdle`.

Choose on substance, not fit

Zero-inflation says some zeros are of a different *kind*; a hurdle says crossing from zero to one is a different *process*. Ask which story your theory tells before letting AIC decide. And check first whether a negative binomial already accommodates the zeros, often it does.

Event History

Durations Are Different (Thursday's Session) ★

Time-to-event outcomes are everywhere in sociology: age at first birth, time to marriage or divorce, job tenure, time to reoffending, organizational survival.

The defining problem: censoring

For many units the clock is still running when observation ends: a respondent aged 35 without a child is not a "no," she is a *not yet*. We observe $\min(T_i, C_i)$ plus an indicator δ_i for whether the event was seen.

- Dropping censored cases selects on the outcome; coding them as event-free forever understates durations. Both bias.
- The fix is this week's principle: write down the probability of exactly what you observed, censoring included.

The Two Building Blocks: Survival and Hazard ★

For a duration T with density $f(t)$:

$$S(t) = \Pr(T > t) \qquad \text{and} \qquad h(t) = \frac{f(t)}{S(t)}.$$

- $S(t)$, the **survival function**: the share still event-free at t. Starts at 1, never rises.
- $h(t)$, the **hazard**: the event *rate* at t among those still at risk, the chance of a first birth at age 30 *given* childless at 30. This conditional rate is what sociological theory is usually about; it is the CEF of the duration world.
- Either determines the other; models are written on the hazard scale.

The Likelihood That Censoring Demands ★

Each unit contributes the probability of what we actually saw:

- event observed at t_i ($\delta_i = 1$): contribute the density $f(t_i)$;
- censored at t_i ($\delta_i = 0$): contribute $S(t_i) = \Pr(T > t_i)$, the probability of lasting past the point where we lost sight of them.

$$\mathcal{L} = \prod_{i=1}^{n} f(t_i)^{\delta_i}\, S(t_i)^{1-\delta_i} \qquad \Longrightarrow \qquad \ell = \sum_i \Big[\delta_i \log f(t_i) + (1-\delta_i)\log S(t_i)\Big].$$

This is the payoff of likelihood thinking: censored units are neither dropped nor mislabelled; each contributes exactly the information it carries. No least-squares analogue can do this.

Kaplan–Meier: $S(t)$ Without a Model ★

Order the observed event times $t_{(1)} < t_{(2)} < \cdots$; let n_j be the number still at risk just before $t_{(j)}$ and d_j the number of events there:

$$\hat{S}(t) = \prod_{j:\, t_{(j)} \le t} \left(1 - \frac{d_j}{n_j}\right).$$

- Read it as a chain of conditional survivals: to be event-free at t you must get past each earlier event time, and $1 - d_j/n_j$ estimates each step.
- Censored units sit in the risk sets n_j while observed, then leave: their partial information is used, then released.
- The step-function plot of $\hat{S}(t)$, by group, is the field's standard descriptive. Look at it before any model.

The Cox Proportional-Hazards Model ★

Put covariates on the hazard, multiplicatively:

$$h(t \mid X_i) = h_0(t)\, \exp(\beta_1 X_{1i} + \cdots + \beta_k X_{ki}).$$

- $h_0(t)$, the **baseline hazard**, is left completely unspecified: no claim about *when* events happen, only about how covariates scale the rate.
- e^{β_j} is a hazard ratio: $e^{\beta_j} = 1.3$ means a 30% higher event rate at every moment, per unit of X_j.
- Estimation ★: the *partial likelihood* compares, at each event time, the unit that failed with everyone still at risk; $h_0(t)$ cancels out of every comparison (Cox 1972).
- The name is the assumption: proportional hazards, a constant ratio between any two units over time. Check it (`cox.zph`) rather than assume it.

Discrete Time: Event History as the Logit You Already Know ★

Sociological clocks are usually discrete (ages, years, waves), and a standard trick reduces event history to this week's tools (Allison 2014):

1. Expand to **person-period** data: one row per unit per period at risk (childless person-years, employed person-months).
2. Outcome = 1 in the period the event occurs, 0 otherwise; a unit's rows stop at its event or its censoring.
3. Run a logit of the outcome on covariates plus period dummies, which act as a nonparametric baseline hazard.

- The coefficients are discrete-time (log) hazard ratios, and time-varying covariates enter for free, updated row by row.
- Censoring is handled by construction: censored units simply stop contributing rows, the likelihood logic from three frames back.

Event History in R, and What to Watch

- `survival::Surv(time, event)` builds the outcome; `survfit()` gives Kaplan–Meier; `coxph()` fits the Cox model; `cox.zph()` checks proportionality.
- Discrete time: build the person-period data, then plain `glm(..., binomial)`. Everything from this week applies, marginal effects included.
- **Watch for**: informative censoring (dropout related to event risk breaks the likelihood above); left truncation (units entering the risk set late); unobserved heterogeneity ("frailty") making hazards look spuriously declining.
- Depth: Allison (2014) is the standard sociological treatment.

Practice

Separation

If some linear combination of the covariates perfectly predicts the outcome, say, every respondent with a graduate degree voted, then the likelihood has no interior maximum. The coefficient marches off to $\pm\infty$.

- Symptoms: enormous coefficients (10, 20, 40) with enormous standard errors; convergence warnings; results that change when you change `maxit`.
- Common with rare outcomes, small samples, and many dummy variables, i.e. very common in practice.
- **Do not** respond by dropping the variable; that discards the strongest signal in the data.
- **Fix** (Zorn 2005): Firth's penalized likelihood, which adds a Jeffreys-prior penalty that guarantees finite estimates and also reduces small-sample bias. `logistf` or `brglm2`.

Assessing Fit

- **Deviance** $= 2(\ell_{\text{saturated}} - \ell_{\text{fitted}})$: the GLM analogue of the residual sum of squares. The saturated term is common to nested models, so *differences* in deviance are exactly LR tests.
- **Pseudo-R^2** (McFadden: $1 - \ell_{\text{full}}/\ell_{\text{null}}$). Interpretable only as a rank ordering, never as "proportion of variance explained." Values of 0.2–0.4 are considered excellent.
- **Classification accuracy**: needs a threshold, and is misleading with rare outcomes, always compare against the null model that predicts the majority class.
- **ROC / AUC**: accuracy across all thresholds. 0.5 is a coin flip, 1.0 is perfect.
- **Predictive distribution checks**: simulate outcomes from the fitted model and compare their distribution to the data. This is the most informative diagnostic and the least often used.

A Practical Protocol ★

1. Choose the distribution from the *outcome*, and the link from convention. Do not agonize over logit versus probit.
2. Fit, then check convergence and look for separation.
3. Check overdispersion for counts; check proportional odds for ordered outcomes.
4. Report average marginal effects or predicted probabilities, with intervals from the delta method or simulation. Coefficient tables belong in an appendix.
5. Never compare raw coefficients across nested logit models or across groups. Compare marginal effects.
6. Use robust or clustered standard errors, exactly as in Week 2, the sandwich applies to `glm` objects too.

Looking Ahead

- We now have both estimation engines: least squares and maximum likelihood. Everything in the rest of the course is one of them, applied to a cleverly chosen quantity.
- Notice what we have *not* done: nothing in Weeks 1–4 says any coefficient is a causal effect. We have been describing conditional distributions.
- **Week 5** changes the question. Potential outcomes and DAGs give us a language for saying what we *want* to estimate, and what would have to be true to get it.

Problem Set 2

Assigned today, due Monday September 28. Covers Weeks 3–4: deriving and coding likelihoods, information and standard errors, and interpreting limited dependent variable models on the GSS.

References

- Ai, Chunrong and Edward C. Norton. 2003. "Interaction Terms in Logit and Probit Models." *Economics Letters* 80(1): 123–129.
- Allison, Paul D. 1999. "Comparing Logit and Probit Coefficients Across Groups." *Sociological Methods & Research* 28(2): 186–208.
- Allison, Paul D. 2014. *Event History and Survival Analysis*, 2nd ed. Sage.
- Berry, William D., Jacqueline H.R. DeMeritt, and Justin Esarey. 2010. "Testing for Interaction in Binary Logit and Probit Models: Is a Product Term Essential?" *American Journal of Political Science* 54(1): 248–266.
- Cox, D.R. 1972. "Regression Models and Life-Tables." *Journal of the Royal Statistical Society, Series B* 34(2): 187–220.
- Hanmer, Michael J. and Kerem Ozan Kalkan. 2013. "Behind the Curve: Clarifying the Best Approach to Calculating Predicted Probabilities and Marginal Effects from Limited Dependent Variable Models." *American Journal of Political Science* 57(1): 263–277.
- James, Gareth et al. 2021. *An Introduction to Statistical Learning*, 2nd ed. Springer. [Ch. 4]

References (cont.)

- King, Gary, Michael Tomz, and Jason Wittenberg. 2000. "Making the Most of Statistical Analyses: Improving Interpretation and Presentation." *American Journal of Political Science* 44(2): 347–361.
- Long, J. Scott. 1997. *Regression Models for Categorical and Limited Dependent Variables*. Sage. [the standard sociological reference]
- Mood, Carina. 2010. "Logistic Regression: Why We Cannot Do What We Think We Can Do, and What We Can Do About It." *European Sociological Review* 26(1): 67–82.
- Zorn, Christopher. 2005. "A Solution to Separation in Binary Response Models." *Political Analysis* 13: 157–170.